

Result

Size

Time

Cycles

SM Frequency

Process

Attributes

Current

582 - batched_cost_kernel

(1024, 1, 1)x(1024, 1, 1)

173.92 us

1,01,730

0 - Tesla T4

584.89 Mhz

[31766] bench_gpu

Summary

Details

Source

Context

Comments

Raw

Session

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GPU Speed Of Light Throughput

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	12.68	Duration [us]	173.92
Memory Throughput [%]	56.31	Elapsed Cycles [cycle]	1,01,730
L1/TEX Cache Throughput [%]	61.75	SM Active Cycles [cycle]	94,229.65
L2 Cache Throughput [%]	51.73	SM Frequency [Mhz]	584.89
DRAM Throughput [%]	27.57	DRAM Frequency [ghz]	4.97

Latency Issue

This workload exhibits low compute throughput and memory bandwidth utilization relative to the peak performance of this device. Achieved compute throughput and/or memory bandwidth below 60.0% of peak typically indicate latency issues. Look at [Scheduler Statistics](#) and [Warp State Statistics](#) for potential reasons.

Key Performance Indicators

Roofline Analysis

The ratio of peak float (fp32) to double (fp64) performance on this device is 32:1. The workload achieved 2% of this device's fp32 peak performance and 0% of its fp64 peak performance. See the [Kernel Profiling Guide](#) for more details on roofline analysis.

Key Performance Indicators

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [cycle]	20,000	# Pass Groups	1
Maximum Buffer Size [Kbyte]	393.22	Dropped Samples [sample]	0

PM Sampling Data

Sampling interval is larger than 10% of the workload duration, which likely results in very few collected samples.

Key Performance Indicators

SM

Average Active Warps Per Cycle	26.4 warp
Total Active Warps Per Cycle	1.06k warp
SM Active Cycles	13.2k cycle
Executed Ipc Active	0.36 inst/cycle

DRAM

DRAM Throughput	100%
DRAM Read Bandwidth	100%
DRAM Write Bandwidth	100%

L1 Cache

Writeback Throughput	1.25k cycle
Hit Rate	100%
Wavefronts (Data)	1.68k

Workload Execution

batched_cost_kernel

No data available. Check PM sampling settings to adjust interval and buffer size.

Compute Workload Analysis

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.31	SM Busy [%]	8.51
Executed Ipc Active [inst/cycle]	0.34	Issue Slots Busy [%]	8.51
Issued Ipc Active [inst/cycle]	0.34		

Low Utilization

Est. Local Speedup: 93.48%

All compute pipelines are under-utilized. Either this workload is very small or it doesn't issue enough warps per scheduler. Check the [Launch Statistics](#) and [Scheduler Statistics](#) sections for further details.

Key Performance Indicators

Pipe Utilization (% of active cycles)

FMA	10.0%
ALU	5.0%
FP64	0.0%
Shared (FP16+Tensor)	0.0%
Tensor (All)	0.0%
Tensor (FP)	0.0%
Tensor (INT)	0.0%

Pipe Utilization (% of peak instructions executed)

LSU	15.0%
FMA	10.0%
XU	5.0%
ALU	5.0%
ADU	2.0%
CBU	2.0%
FP16	0.0%
FP64	0.0%
TEX	0.0%
Tensor (FP)	0.0%
Tensor (INT)	0.0%
Uniform	0.0%

Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	87.62	Mem Busy [%]	51.73
L1/TEX Hit Rate [%]	10.65	Max Bandwidth [%]	56.31
L2 Hit Rate [%]	88.31	Mem Pipes Busy [%]	12.68
L2 Compression Input Sectors [sector]	n/a	L2 Compression Success Rate [sector]	n/a

L1TEX Global Load Access Pattern

Est. Speedup: 39.42%

The memory access pattern for global loads from L1TEX might not be optimal. On average, only 9.6 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the [Source Counters](#) section for uncoalesced global loads.

Key Performance Indicators

Memory Chart

Values: Transfer Size Inactivity: Greyed Out

Scheduler Statistics

Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	6.91	No Eligible [%]	91.38
Eligible Warps Per Scheduler [warp]	0.14	One or More Eligible [%]	8.62
Issued Warp Per Scheduler	0.09		

Issue Slot Utilization

Est. Local Speedup: 43.69%

Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 11.6 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 8 warps per scheduler, this workload allocates an average of 6.91 active warps per scheduler, but only an average of 0.14 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, avoid possible load imbalances due to highly different execution durations per warp. Reducing stalls indicated on the [Warp State Statistics](#) and [Source Counters](#) sections can help, too.

Key Performance Indicators

Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined values.

Warp Cycles Per Issued Instruction [cycle]	80.19	Avg. Active Threads Per Warp	31.99
Warp Cycles Per Executed Instruction [cycle]	80.51	Avg. Not Predicated Off Threads Per Warp	31.17

Long Scoreboard Stalls

Est. Speedup: 43.69%

On average, each warp of this workload spends 54.1 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 67.4% of the total average of 80.2 cycles between issuing two instructions.

Key Performance Indicators

Warp Stall

Check the [Warp Stall Sampling \(All Samples\)](#) table for the top stall locations in your source based on sampling data. The [Kernel Profiling Guide](#) provides more details on each stall reason.

Key Performance Indicators

Instruction Statistics

Statistics of the executed low-level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	12,78,183	Avg. Executed Instructions Per Scheduler [inst]	7,988.64
Issued Instructions [inst]	12,83,303	Avg. Issued Instructions Per Scheduler [inst]	8,020.64

FP32 Non-Fused Instructions

Est. Speedup: 2.04%

This kernel executes 98304 fused and 163840 non-fused FP32 instructions. By converting pairs of non-fused instructions to their [fused](#), higher-throughput equivalent, the achieved FP32 performance could be increased by up to 31% (relative to its current performance). Check the Source page to identify where this kernel executes FP32 instructions.

Key Performance Indicators

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

Key Performance Indicators

NVLink Tables

Detailed tables with properties for each NVLink.

Key Performance Indicators

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

Key Performance Indicators

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	1,024	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	16	Static Shared Memory Per Block [byte/block]	0
Block Size	1,024	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	10,48,576	Driver Shared Memory Per Block [byte/block]	0
Waves Per SM	25.60	Shared Memory Configuration Size [Kbyte]	32.77
Uses Green Context	0	Stack Size	1,024
# SMs [SM]	40	# TPCs	20
Enabled TPC IDs	all		-

Key Performance Indicators

Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	4
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	16
Achieved Occupancy [%]	85.68	Block Limit Warps [block]	1
Achieved Active Warps Per SM [warp]	27.42	Block Limit SM [block]	16

Achieved Occupancy

Est. Speedup: 14.32%

The difference between calculated theoretical (100.0%) and measured achieved occupancy (85.7%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the [CUDA Best Practices Guide](#) for more details on optimizing occupancy.

Key Performance Indicators

GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	94,229.65	Average L1 Active Cycles [cycle]	94,229.65
Average L2 Active Cycles [cycle]	1,42,013.72	Average SMSP Active Cycles [cycle]	93,020.93
Average DRAM Active Cycles [cycle]	2,38,120	Average MC Channel Active Cycles [cycle]	n/a
Total SM Elapsed Cycles [cycle]	41,33,208	Total L1 Elapsed Cycles [cycle]	41,33,208
Total L2 Elapsed Cycles [cycle]	47,57,056	Total SMSP Elapsed Cycles [cycle]	1,65,32,832
Total DRAM Elapsed Cycles [cycle]	69,09,952	Total MC Channel Elapsed Cycles [cycle]	n/a

Key Performance Indicators

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	1,31,177	Branch Efficiency [%]	99.94
Branch Instructions [inst]	0.10	Avg. Divergent Branches [branches]	0.13

Uncoalesced Global Accesses

Est. Speedup: 63.69%

This kernel has uncoalesced global accesses resulting in a total of 1835008 excessive sectors (67% of the total 2752512 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The [CUDA Programming Guide](#) has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
instance_update.cu:50 (0x12af2325a400 in batched_cost_kernel)	7,86,432	43
instance_update.cu:51 (0x12af2325a400 in batched_cost_kernel)	7,86,432	43
instance_update.cu:48 (0x12af2325a390 in batched_cost_kernel)	1,31,072	7
instance_update.cu:47 (0x12af2325a380 in batched_cost_kernel)	1,31,072	7
instance_update.cu:53 (0x12af2325a580 in batched_cost_kernel)	0	0

To customize your report even further, you might want to learn about [custom sections](#) and [writing your own rules](#). You might also want to consider [adding individual metrics](#).

You can ask [Night Copilot](#) questions about this report.